Georgia Zhang

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Education

University of Toronto Toronto, ON
BASc., Computer and Electrical Engineering + ***PEY Co-op****; GPA: 3.73/4.0, Dean's List* *Sep. 2024 – May 2029*

- Relevant Coursework: Digital Systems, Computer Organization, Hardware Design and Communication, Digital Electronics, Computer Architecture, Operating Systems

Experience

Embedded Systems Student Intern May 2026 – Aug. 2026
Network Research Lab, University of Toronto *Toronto, ON*

- Implemented concurrent **FreeRTOS** tasks for sensor collection and cloud transmission, coordinating tasks with queues and sleep states to maintain responsiveness and reduce power use.
- Designed and built a **4-layer electrochemical breakout board** with the **LMP91000** and **ADS122C04 24-bit ADC** to measure **nanoamp currents** from electrochemical sensors.
- Fixed ESP32 boot failures from a **GPIO12 strapping pin conflict** by delaying software serial initialization 10s.
- Fixed ESP32 brownouts from **LDO current limits** during modem TX by powering the modem from battery.

Firmware Developer May 2026 – Present
University of Toronto Aerospace Team (UTAT) – Space Systems *Toronto, ON*

- Configured **MCUboot** on **Zephyr RTOS** for **STM32G431RB**, fixing DTS/flash errors for firmware updates.
- Used **SSH** and the **UTAT VPN** to remotely flash STM32 firmware through a Raspberry Pi using PyOCD.

VP of Partnerships May 2026 – Present
University of Toronto Robotics Association (UTRA) *Toronto, ON*

- Created **sponsorship packages** and technical reports to communicate needs and build industry partnerships.
- Coordinated parts, components, and funding across **Combat, ART, PacBots, RoboSoccer, and SUMO**.

SUMO Bot Team Project Manager May 2025 – May 2026
University of Toronto Robotics Association (UTRA) *Toronto, ON*

- Led firmware and hardware setup sessions for **30+ teams** of 3 engineers building autonomous **SUMO robots**.
- Ran **Arduino C++** and hardware workshops covering sensors, wiring, and soldering for new team members.
- Coordinated a SUMO tournament with **York, Ontario Tech, TMU, and McMaster** to manage event logistics.

Projects

Digital Logic Simulator & Compiler Mar. 2026 – Apr. 2026

- Built a bare-metal **C** digital logic simulator on **NIOS-V** for executing combinational and sequential circuits on the **DE1-SoC**.
- Implemented **VGA** graphics and 90-degree wire routing to visualize circuit structure and live logic states.
- Developed a **C#** compiler to translate text-based circuit definitions into binary instructions for **FPGA memory**.
- Integrated **PS/2** keyboard input to modify circuit inputs during execution and generate truth tables.

Vehicle Lighting Control Board Dec. 2025 – Mar. 2026

- Designed a **KiCad** PCB with 12V, 5V, and 3.3V rails and an **ESP32** for vehicle lighting control.
- Implemented **C++** firmware to read IMU orientation and trigger lights when rollover thresholds were exceeded.
- Added **CAN bus** communication to exchange vehicle state with other control boards.

Voice-Controlled Hardware Game Oct. 2025 – Dec. 2025

- Built a **Verilog** hardware game that processes microphone samples into a loudness signal, mapping voice amplitude to player movement on a **VGA** display.
- Redesigned **VGA** rendering from pixel-by-pixel to column-based drawing, eliminating visual streaking.
- Wrote **Verilog** testbenches in **ModelSim** to verify modules and debug obstacle generation and movement.

Technical Skills

Languages: C/C++, Python, C#, Verilog, Assembly
Embedded Systems: STM32, ESP32, ATtiny, NIOS-V, FreeRTOS, Zephyr RTOS, MCUboot, CAN, I2C, UART, LoRa
Hardware & Debugging: KiCad, Oscilloscope, Multimeter, Soldering, GDB
Software & Tools: Git, Linux, AWS IoT, InfluxDB, Grafana